

Third Normal Form (3NF) — Relational Database Design

Definition: A relation is in **Third Normal Form (3NF)** if it is in **2NF** and **no non-prime attribute is transitively dependent on the primary key**. For every non-trivial functional dependency $X \rightarrow Y$, either **X is a super key** or **Y is a prime attribute**.

1. Detailed Technical Explanation

Transitive Dependency Concept:

A transitive dependency exists if $A \rightarrow B$ and $B \rightarrow C$ hold, which implies $A \rightarrow C$ through non-key attribute B.

```
[ Primary Key A ] -----> [ Non-Prime Attribute B ]
                             |
                             v
                        [ Non-Prime Attribute C ]
                    (Transitive Dependency!)
```

Conversion Example: 2NF to 3NF

Relation EMP_DEPT(Emp_ID, Emp_Name, Dept_ID, Dept_Name, Dept_Head)

- **Candidate Key:** Emp_ID
- **Functional Dependencies:**
 1. Emp_ID $\rightarrow$ Emp_Name, Dept_ID
 2. Dept_ID $\rightarrow$ Dept_Name, Dept_Head (Transitive Dependency: Dept_ID is not a super key, Dept_Name is non-prime)

Decomposition into 3NF:

We decompose EMP_DEPT into 2 distinct tables:

1. **EMPLOYEE (Emp_ID, Emp_Name, Dept_ID)**
 - Primary Key: Emp_ID
 - Foreign Key: Dept_ID
2. **DEPARTMENT (Dept_ID, Dept_Name, Dept_Head)**
 - Primary Key: Dept_ID

2. Core Concepts & Memory Keywords

- **Transitive Dependency:** Non-key attribute determining another non-key attribute.
- **3NF Formal Condition:** For all $X \rightarrow Y$:
 1. $X \rightarrow Y$ is trivial ($Y \subseteq X$), OR

2. X is a Super Key, OR
 3. Y is a Prime Attribute (part of a candidate key).
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3. Must-Write Points for Exams

- 3NF guarantees **lossless-join decomposition** and **dependency preservation**.
 - 3NF removes insertion and deletion anomalies associated with transitive attribute relationships.
 - Most practical commercial enterprise databases are normalized up to 3NF.
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4. Quick Recall Flow

2NF Relation -> Check X -> Y -> Ensure X is Super Key OR Y is Prime -> Remove Transitive Dependencies -> 3NF Achieved